Huawei Access Node SOP: Resolving Optical Transport Failure (ALARM_3370) on HEB-ACC-75 at Hebbal Edge

Node: HEB-ACC-75 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_3370

Description

This SOP outlines the diagnostic and resolution procedures for resolving a Major Optical Transport Failure (ALARM_3370) on the Huawei Access Node (HEB-ACC-75) at the Hebbal Edge site. The failure domain encompasses various causes such as SFP/XFP degradation, fiber cuts, and Rx/Tx power loss, which can be attributed to the underlying hardware and protocol mechanics of the Huawei Access Node. The device's Optical Transport Network (OTN) module relies on the Huawei proprietary OTU (Optical Transport Unit) protocol, which transports client signals over optical fibers. The OTU protocol is responsible for framing, multiplexing, and demultiplexing client signals onto the fiber optic transmission medium. A failure in this domain can be caused by physical damage to the fiber optic cables, SFP/XFP degradation, or incorrect OTU protocol configuration. This SOP provides a step-by-step guide for diagnosing and resolving such failures, ensuring minimal downtime and optimal network performance.

Pre-requisite Checks

- check: Verify the site location and ensure that the Hebbal Edge site is accessible for remote hands support., status: Completed

- check: Check the redundant link status to ensure that it is operational and can take over in case of a failure., status: Completed

- check: Perform a visual inspection of the fiber optic cables and SFP/XFP modules to identify any physical damage or issues., status: Completed

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-transport interface, Expected Output: Interface Name Admin State Oper State, Description: This command displays the current state of the optical transport interface, including the admin and oper states. It helps to identify any issues with the interface, such as fiber cuts or SFP/XFP degradation.
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```
Command: display optical-transport signal, Expected Output: Signal Name Admin State Oper State Power, Description: This command displays the current state of the optical transport signal, including the admin and oper states, as well as the power level. It helps to identify any issues with the signal, such as Rx/Tx power loss.

Command: display optical-transport otu, Expected Output: OTU Name Admin State Oper State Protocol, Description: This command displays the current state of the OTU protocol, including the admin and oper states, as well as the protocol used. It helps to identify any issues with the OTU protocol configuration.
```

Resolution Steps

Step 1: Step: Replace SFP/XFP module, Command: replace optical-transport interface, Description: Replace the SFP/XFP module with a new one to resolve SFP/XFP degradation issues.

Step 2: Step: Inspect and repair fiber optic cables, Command: display optical-transport interface, Description: Inspect the fiber optic cables for any physical damage and repair them if necessary to resolve fiber cuts issues.

Step 3: Step: Adjust OTU protocol configuration, Command: modify optical-transport otu, Description: Adjust the OTU protocol configuration to resolve issues with the OTU protocol configuration.

Post-Check Verification

```
Command: display optical-transport interface, Expected Output: Interface Name Admin State Oper State, Description: Verify that the optical transport interface is functioning properly and that the admin and oper states are correct.

Command: display optical-transport signal, Expected Output: Signal Name Admin State Oper State Power, Description: Verify that the optical transport signal is functioning properly and that the admin and oper states, as well as the power level, are correct.

Command: display optical-transport otu, Expected Output: OTU Name Admin State Oper State Protocol, Description: Verify that the OTU protocol is functioning properly and that the admin and oper states, as well as the protocol used, are correct.
```

Roll-back Procedures

- Step: Restore original SFP/XFP module, Command: undo replace optical-transport interface, Description: Restore the original SFP/XFP module to resolve any issues caused by replacing it.

- Step: Restore original OTU protocol configuration, Command: undo modify optical-transport otu, Description: Restore the original OTU protocol configuration to resolve any issues caused by adjusting it.